5) Week 13: Matrix operations

5.1) For the matrices $A \in \mathbb{R}^{5\times 2}$ and $B \in \mathbb{R}^{2\times 3}$, is the matrix multiplication AB a valid operation? If so, what is the shape of AB?

Yes, it is a valid operation.

$$\underset{A}{\begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix}} \underset{B}{\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}} = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \in \mathbb{R}^{5\times 3}$$

5.2) What spaces do A, x, and b belong to?

$$A = \begin{bmatrix} 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}, \; b = (8, 0)\,, \; x = \begin{bmatrix} 0 & 3 & 1 \end{bmatrix}$$

- ○ $A \in \mathbb{R}^{3\times 2}$, $b \in \mathbb{R}^2$, $x \in \mathbb{R}^3$
- ○ $A \in \mathbb{R}^{2\times 3}$, $b \in \mathbb{R}^3$, $x \in \mathbb{R}^2$
- ● $A \in \mathbb{R}^{2\times 3}$, $b \in \mathbb{R}^2$, $x \in \mathbb{R}^3$
- ○ $A \in \mathbb{R}^{2\times 3}$, $b \in \mathbb{R}^2$, $x \in \mathbb{R}^2$

One can also say $x \in \mathbb{R}^{1\times 3}$

5.3) Given the matrix $A \in \mathbb{R}^{19\times 23}$, what space of matrices could B belong to in order for the product BA to be well-defined? Choose all that apply.

- ● $\mathbb{R}^{23\times 19}$
- ○ $\mathbb{R}^{23\times 23}$
- ○ $\mathbb{R}^{46\times 23}$
- ○ $\mathbb{R}^{19\times 23}$

$$\overset{19}{\begin{bmatrix} B \in \mathbb{R}^{a\times 19} \end{bmatrix}} \overset{23}{\begin{bmatrix} A \in \mathbb{R}^{19\times 23} \end{bmatrix}} \; 19$$

5.4) Given the following matrices G and H:

$$G = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad H = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$$

a) What is the value of HG?

b) What is the dimension of GH?

a) $$HG = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$= 2 \times 2 + 3 \times (-1) + 4 \times 3$$

$$= 4 - 3 + 12 = 13$$

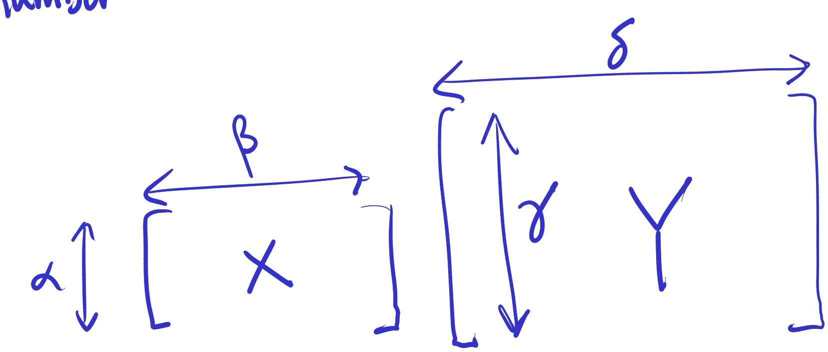
b) $G \in \mathbb{R}^{3 \times 1}$, $H \in \mathbb{R}^{1 \times 3}$ $\Rightarrow$ $GH \in \mathbb{R}^{3 \times 3}$

5.5) Given matrices $A \in \mathbb{R}^{2\times 4}$ and $B \in \mathbb{R}^{\alpha\times\beta}$

a) What should α be so that AB is well-defined?

b) What should β be so that BA is well-defined?

- In general, with $X \in \mathbb{R}^{\alpha\times\beta}$ and $Y \in \mathbb{R}^{\gamma\times\delta}$ we can multiply XY only if $\beta = \gamma$.
- That is, the number of columns of X must equal the number of rows in Y.



- With $A \in \mathbb{R}^{2\times 4}$, $B \in \mathbb{R}^{\alpha\times\beta}$

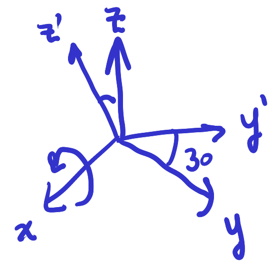
a) AB is defined only if $\alpha = 4$

b) BA is defined only if $\beta = 2$

6) Week 13: 3D Rotations and Projections

6.1) Let $p = (1, 2, 1)$ represent the position of a penny in 3D space relative to the (x, y, z) coordinate frame. Find the position of the penny in the coordinate frame (x'', y'') obtained by applying the following transformations to (x, y, z):

1. Rotate (x, y, z) by 30° about the x axis. This produces the (x', y', z') coordinate frame.
2. Perform a projection of the (x', y', z') coordinate frame onto (x'', y'') in which the z' coordinate becomes the y'' corrdinate ($z' \to y''$), and the y' coordinate becomes the x'' coordinate ($y' \to x''$).



1.

$$\begin{bmatrix} P_{x'} \\ P_{y'} \\ P_{z'} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30 & \sin 30 \\ 0 & -\sin 30 & \cos 30 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & 1/2 \\ 0 & -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3} + \frac{1}{2} \\ -1 + \frac{\sqrt{3}}{2} \end{bmatrix}$$

2.

$$\begin{bmatrix} P_{x''} \\ P_{y''} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} P_{x'} \\ P_{y'} \\ P_{z'} \end{bmatrix} = \begin{bmatrix} \sqrt{3} + 1/2 \\ -1 + \sqrt{3}/2 \end{bmatrix}$$

6.2) Let $p = (p_x, p_y, p_z)$ represent a point in 3D space. Let

$$Z = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a matrix used to rotate the coordinate frame by $\beta = 45°$ around the z axis. Which of the following is the correct formula for determining p's coordinates with respect to the rotated coordinate frame?

- ○ pZ
- ● Zp
- ○ $p + Z$

Note: This is following the convention of this class, set out in lecture, in which $p = (p_x, p_y, p_z)$ should be arranged as a column vector and multiplied on the right side of the rotation matrix Z.

If you have learned a different convention in another class, you must still do it this way on the final.

7) Week 13: Systems of linear equations

7.1) Express the following system of equations in matrix form.

$$\begin{aligned} 1 - x_2 &= 3\,x_3 \\ x_1 &= 7 - x_2 \\ x_1 + x_2 + x_3 &= 0 \end{aligned}$$

Organize variables on the left and constants on the right.

$$\begin{cases} -x_2 - 3x_3 = -1 \\ x_1 + x_2 = 7 \\ x_1 + x_2 + x_3 = 0 \end{cases}$$

Then with $X = (x_1, x_2, x_3)$

$$\underbrace{\begin{bmatrix} 0 & -1 & -3 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} -1 \\ 7 \\ 0 \end{bmatrix}}_{b}$$

7.2) A system of linear equations $Ax = b$ is known to have infinitely many solutions. What can be said about the rank of A?

The rank of A must be less than n, the number of unknowns.

7.3) Express the following system of equations in matrix form.

$$\begin{aligned} x_1 + x_2 &= 1 \\ x_1 + x_3 &= 2 \\ x_2 + x_3 &= 3 \\ 2x_1 + 2x_2 + 2x_3 &= 4 \end{aligned}$$

With $x = (x_1, x_2, x_3)$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

7.4) A system of linear equations is given in matrix form $Ax = b$, with $A \in \mathbb{R}^{2\times 5}$.

a) What is the dimension of x?

b) What is the dimension of b?

c) How many solutions are there to the system of equations if both rank(A) and rank(A|b) equal 2?

$$\begin{bmatrix} \vdots & \vdots & A & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \vdots \\ x \\ \vdots \end{bmatrix} = \begin{bmatrix} b \end{bmatrix}$$

a) $x \in \mathbb{R}^5$ ... $n = 5$

b) $b \in \mathbb{R}^2$

c) $\text{rank}(A) = \text{rank}(A|b) = 2 < n.$

$\Rightarrow$ infinitely many solutions.

7.5) The rank of A is 3 and the rank of $A|b$ is 4. How many solutions are there to this system of equations $Ax = b$?

rank $(A) = 3$

rank $(A|b) = 4$

rank $(A|b) >$ rank $(A) \Rightarrow 0$ solutions.

7.6) You are given the matrix equation $Ax = b$, where b is not in the span of A. How many possible x satisfy this equation?

Since $b \notin$ span (A), we have rank $(A|b) >$ rank (A)

$\Rightarrow$ no solutions.